The loop isn't a gate, it's a gain

BLAKE

There's a governance idea going around that says an AI system should be required to run a correction loop, disclose a calibrated confidence, and flag uncorrected deviations. Good idea. I think one piece of it is built wrong.

The correction loop is written as a gate. The system passes or fails. But the thing the idea borrows from, precision-weighting in predictive processing, is not a gate. It's a gain. Prediction error gets multiplied by a weight before it updates anything, and that weight is set by how reliable the system takes the signal to be. A gain can be too low, too high, or right. It's a dial, not a switch.

That difference is not philosophical. Take the failure cases the framework was built for. In psychosis the loop is running. The gain on prediction error is wrong. Same story in the psychedelic case. If your rule is pass/fail, you certify a system compliant while its correction runs at the wrong gain. You've measured the presence of the loop, not its calibration. That is the exact failure the rule exists to catch.

I have a small dataset that makes the point in a different domain.

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32 people listened to the same 40 one-minute music videos.
Pulse and respiration recorded throughout. Variance decomposition:

heart rate person 89.3% song 0.5% person x song 10.2%
respiration person 38.5% song 2.3% person x song 59.3%
liking song 34% person x song 58%

Not one of the 40 songs moved hearts in a shared direction.
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Same input. Different bodies. Wildly different responses. The stimulus does not determine the outcome. The person does.

If you wrote a governance rule off that result, you would not write "the stimulus passes or fails." You would write: **state the gain, and disclose whose body set it.**

That's calibrated confidence disclosure, one level down. The confidence signal is the gain. Calibration is whether the gain is right. And an uncalibrated gain is worse than no gain at all, because it borrows precision it does not have.

The same disease shows up in the numbers people quote. I checked one from a post I read: "cut one leading model's hallucination rate nearly in half, accuracy almost unchanged." It traces to one model, one benchmark, reported on a blog. The same model class gets called 3 to 8 percent, 16 to 33 percent,

and 88 percent in sources I pulled, because each measured a different task. A rate without its protocol is an uncorrected deviation. The framework, running on its own sentence.

So: keep the correction loop. Add a gain, not just a gate. And put a denominator on every rate, including your own.

DEAP reanalysis, 32 subjects x 40 music videos. Pulse and respiration. Reproduced independently twice. Numbers and scripts on request.